Robustness of Interpretable Temporal and Spectral Features to Real-World Background Noise in Environmental Sound Classification

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GitHub repository: github.com/IrisHe177/elec5305-project-540662177

GitHub Pages: irishe177.github.io/elec5305-project-540662177

Video link: https://drive.google.com/file/d/1HSP13Enzi-fcHtOALvW61lwVB0cJMc0R/view?usp=drive_link

Abstract

Environmental sound classifiers are usually evaluated on clean recordings but used in noisy places. How does real background noise degrade ESC-10 classification based on interpretable temporal and spectral features? How does it compare with a pretrained deep network? Piczak's MFCC + ZCR baseline is reproduced under official folds with nested SVM tuning; this is compared to temporal-only, spectral-only and combined descriptor sets and YAMNet embeddings classified by same SVM. Every test clip is mixed with six DEMAND noises at 20 to −5 dB.

On clean data temporal features beat spectral features (63.0% vs 55.0%), six descriptors match MFCCs (69.3% vs 69.0%), their combination 73.0% and YAMNet 96.5%. In noise clean-trained MFCC models fail at even 20 dB (26.9%). This is because log compression of near empty bands and digitally silent frames moves MFCCs about ten times farther than power-weighted descriptors; the descriptors degrade predictably with noise spectrum, YAMNet is far more robust (63.6% at 0 dB) with semantically plausible errors. Multi-condition training lifts the combined handcrafted model from 25.3% to 66.8% at 0 dB and to 53.8% for noise types unseen in training at a clean-data cost less than two points.

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1 Introduction

Environmental sound classification (ESC) is used in acoustic monitoring, smart devices and assistive systems [2]. In all these settings the sound of interest is recorded together with background noise: traffic, people talking, kitchen clatter, running water or a car cabin. Benchmark results are almost always reported on clean clips so a classifier can be accurate on the benchmark but fail where it is used.

The proposal was to compare temporal features, spectral features and their combination for ESC with conventional classifiers. In the progress report we reproduced Piczak [3] MFCC + zero-crossing-rate (ZCR) baseline on ESC-10 and showed that adding six interpretable descriptors increased SVM accuracy from 64.8% to 73.5%. Feedback at that stage wanted three things: a comparison with state-of-the-art deep network; a much deeper study of how different types of background noise affect performance (e.g. DEMAND database); more interpretation of results. The final project combines the proposal and feedback into one research question.

> ***Research question.*** *How do different types of real-world background noise degrade ESC-10 classification based on interpretable temporal and spectral features, and how does this compare with a pretrained deep audio network?*

Following the course guidance to reproduce an existing approach and make a small modification, the study:

1. reproduces the Piczak MFCC + ZCR baseline under the official ESC-10 folds;
2. evaluates temporal-only, spectral-only, descriptor and combined feature sets, as planned in the proposal;
3. adds a pretrained deep reference (YAMNet embeddings) classified with the same SVM;
4. mixes every test clip with six real DEMAND noise types at five SNRs;
5. tests one modification aimed at the weakness this reveals: multi-condition (noise-augmented) training, including noise types never seen in training.

Throughout, the interpretable features are used to explain why performance changes, not only to measure it.

2 Literature Review

2.1 Handcrafted features for audio classification

Short-time descriptors computed from the waveform and the short-time Fourier spectrum have a long history in audio classification. Tzanetakis and Cook [1] used spectral centroid, rolloff, flux, zero crossing rate and MFCCs to classify musical genres; Peeters [8] catalogued a large set of such descriptors with precise definitions. For environmental sounds Chu et al. [5] argued that MFCCs alone miss important timefrequency structure of non-speech sounds and complementary features improve recognition.

Piczak [3] introduced the ESC-50 dataset and its 10 class subset ESC-10 with a baseline summarising frame-level MFCCs and ZCR by mean and standard deviation. His random forest achieved 72.7% on ESC-10 (human accuracy was 95.7%). Cheap and interpretable are good, but MFCC were designed for speech [7], and behaviour under non-speech background noise is rarely analysed feature by feature.

2.2 Deep and pretrained audio models

Convolutional networks trained on log-mel spectrograms soon outperformed handcrafted baselines. Piczak's CNN [4] reached about 80% on ESC-10 with data augmentation and Salamon and Bello [6] showed that augmentation is needed when data are scarce. Large scale pretraining changed the field further. AudioSet [11] provided two million labelled YouTube clips on which VGGish style CNNs [12], YAMNet (a MobileNet-v1 trained on AudioSet and distributed with MATLAB's Audio Toolbox), PANNs [13] transformers AST [14] and BEATs [15] were trained. On ESC-50 AST reaches 95.7% and BEATs 98.1%, far above handcrafted features [3].

The price is model size, computation, dependence on large external training data and much weaker interpretability. The study uses YAMNet as deep reference. It is in the same family of AudioSet-pretrained models as current state of the art; it runs inside the same MATLAB pipeline folds and noisy mixtures as handcrafted features. With the same SVM it reaches 96.5% on clean ESC-10 (Section 4.1) which is just above human accuracy so a stronger model has little room to improve on clean data; the question is how does it behave in noise?

2.3 Robustness to background noise

Dennis et al. [9] showed that MFCC-based classifiers of sound events trained on clean data degrade sharply in mismatched noise and proposed more robust spectrogram image features. McLoughlin et al. [10] found that training on noisy data and using deep networks both improved robustness to noise. Training on data corrupted by several noise types at several SNRs (multi-condition training) is standard from noisy speech recognition (Aurora-2 [16]). Realistic noise for such experiments can be obtained in DEMAND [17], which provides multichannel recordings of 18 everyday environments.

What is less common to report is a feature-level explanation of why a given representation fails under a given noise. A direct comparison of interpretable descriptors with a pretrained deep network under the same noise conditions is also rare. This study addresses that gap.

2.4 Tools used and concepts learned

The implementation uses MATLAB R2024b with Audio Toolbox (mfcc, spectralSpread, spectralFlatness, spectralRolloffPoint, yamnetPreprocess, audioPretrainedNetwork), Signal Processing, Statistics and Machine Learning, and Deep Learning Toolboxes. Several concepts were new in this project:

- **Nested cross-validation** keeps hyperparameter selection inside the training folds, so test accuracy is not optimistically biased [18].
- **McNemar's test** compares two classifiers on the same test items using only the items they disagree on [19].
- **Transfer learning with frozen embeddings** reuses a pretrained network as a fixed feature extractor.
- **SNR for intermittent sounds** has to be defined on the active part of a clip when the clip contains silence (Section 3.4).

3 Methods

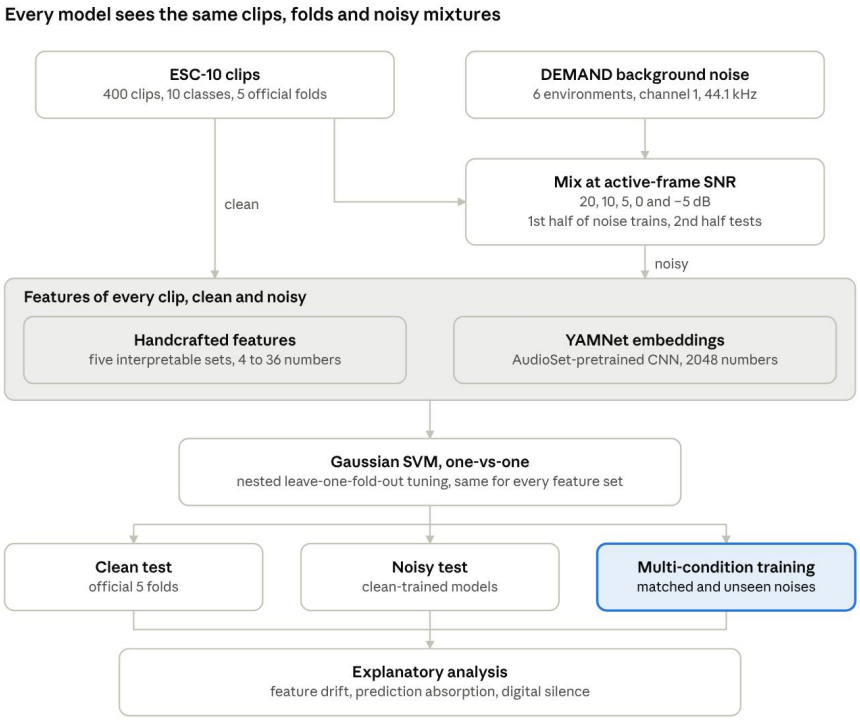


Figure 1. Experimental pipeline. Every feature set sees the same clips, folds and noisy mixtures; the highlighted box is the modification tested (multi-condition training).

3.1 Data and evaluation protocol

ESC-10 contains 400 five-second clips (44.1 kHz, mono) from 10 balanced classes: dog, rooster, rain, sea waves, crackling fire, crying baby, sneezing, clock tick, helicopter and chainsaw. The official folds are used throughout: 320 training and 80 test clips per fold; clips cut from the same source recording never cross folds. Every clip gets exactly one held-out prediction per model.

An important property of the data is that 10.1% of all analysis frames are digital silence (all zeros) because short events were padded to five seconds. Silence is concentrated in three classes: on average 40.9% of frames for sneezing, 35.8% for rooster and 15.6% for dog against about 0% for rain, sea waves and chainsaw.

3.2 Feature sets

All handcrafted features are computed from 2048-sample periodic Hann frames (46.4 ms) with a 512-sample hop, a 2048-point FFT, reflection padding (431 frames per clip) and no amplitude normalisation, the same settings as in the progress report. Each frame-level feature is summarised per clip by its mean and population standard deviation, giving the six feature sets in Table 1.

Table 1. Feature sets compared. All sets use the same classifier and the same folds.

Feature set	Content	Dims	Role
Temporal	RMS energy, ZCR	4	temporal features only
Spectral	spectral centroid, spread, flatness, 90% rolloff	8	spectral features only

Feature set	Content	Dims	Role
Descriptors	temporal + spectral	12	all six interpretable descriptors
MFCC baseline	12 MFCCs (C1–C12) + ZCR	26	reproduced Piczak baseline [3]
Combined	MFCC baseline + six descriptors	36	combined handcrafted set
YAMNet	YAMNet embeddings	2048	pretrained deep reference

For YAMNet, each clip is resampled to 16 kHz and cut into nine 0.96-s log-mel patches, the network's standard input. The 1024-dimensional output of its global-average-pooling layer is taken for each patch and summarised by mean and standard deviation, exactly like the handcrafted features. Using the same classifier for every feature set isolates the effect of the representation.

3.3 Classifier and statistical testing

Every feature set is classified by a one-versus-one Gaussian SVM on features standardised with training fold statistics. Hyperparameters are determined by nested cross validation: for each outer test fold the four training folds form an inner loop of leave-one-fold-out over the box constraint $C \in \{0.1, 1, 10, 100, 1000\}$ and kernel scale $\sqrt{d} \times \{0.5, 1, 2, 4\}$ (where d is the dimension of features). Using whole official folds in the inner loop ensures clips cut from the same source recording are on one side of every split; otherwise a random partition would not guarantee that.

For comparison with the progress report its untuned setting ($C = 1$, kernel scale $\sqrt{26}$) is also reported. Accuracy and macro-F1 are calculated on the 400 pooled out-of-fold predictions. Paired comparisons use exact two-sided McNemar test on clips that one model gets right and other wrong [19].

3.4 Background noise and mixing

Six DEMAND recordings [17], one from each of its six categories were selected for contrasting acoustic properties: car interior (TCAR), street traffic (STRAFFIC), cafeteria (PCAFETER), kitchen (DKITCHEN), river (NRIVER) and meeting room (OMEETING). The DEMAND codes in brackets identify the figure panels. Channel 1 of each 48 kHz recording (300 s) is resampled to 44.1 kHz. First half of each recording is used only for noisy training data and second half only for noisy test data so no noise sample is shared between training and testing.

Each noise is characterised by its long-term average spectrum (LTAS) and by five numbers:

- LTAS centroid
- LTAS flatness
- share of power below 500 Hz
- share of power above 8 kHz (YAMNet sees only 0–8 kHz)
- standard deviation of its frame level in dB, a measure of non-stationarity

A test clip x is mixed with a 5-s noise segment n as $y = x + g \cdot n$, with g chosen to reach the target SNR (20, 10, 5, 0 and −5 dB):

$$g = \sqrt{\frac{P_x}{P_n \cdot 10^{\mathrm{SNR}/10}}}, \qquad P_x = \text{mean power of the active frames of } x$$

Because of the digital silence described above, the signal power is measured only on active frames (within 40 dB of the loudest frame), not on the whole clip. Otherwise a mostly silent sneeze would be tested at a much lower true SNR than a continuous rain recording at the same nominal SNR. Only the

noise is scaled. Each clip's noise segment is drawn once per noise type with a fixed seed and reused at every SNR, and all models see identical mixtures, so every comparison is paired.

3.5 Robustness test and the modification

In robustness test the clean-trained, nested-tuned fold models are applied unchanged to the noisy test clips.

The modification is multi-condition training in which the classifier is trained on clean and noisy versions of the training clips. For every clip one noisy copy per noise type is made from the training half of the noise recordings at an SNR drawn from {20, 10, 5, 0} dB (−5 dB never seen in training). A training fold then contains its clean clips plus noisy copies of those same clips. Two settings are evaluated:

- **matched**: copies from all six noise types;
- **unseen**: copies from the other five noise types only, so the test noise is new to the classifier.

The hyperparameters chosen on clean data are kept, so any change is due to the training data alone.

3.6 Explanatory analyses

- **Feature drift:** how far each feature moves from its clean to its noisy value for the same clip, in units of its within-class standard deviation, for single descriptors and for feature families (means and standard deviations of the MFCCs and of the descriptors).
- **Prediction absorption:** the share of noisy clips assigned to each class, which shows whether errors spread out or collapse onto a few classes.
- **Digital silence:** how often a clip that is correct when clean becomes wrong in noise, split by whether more than 25% of its frames are silent.
- **Cost:** feature dimension, learned parameters and median extraction time per clip.

3.7 Reproducibility

The experiment is implemented as seven MATLAB Live Scripts running in order (from an environment check through clean evaluation, noise mixing and multi-condition training to the final analysis). All random choices use fixed seeds. The noisy features are saved with the results so the analysis can be re-run without downloading the noise database. Code, results and run instructions are in the GitHub repository.

4 Results

4.1 Clean ESC-10: temporal vs spectral features, the baseline and the deep reference

Table 2 answers the comparison planned in the proposal. With nested tuning, the four temporal statistics (63.0%) beat the eight spectral statistics (55.0%; McNemar $p = 0.005$). Together the six descriptors (69.3%) match the baseline MFCC (26 dimensions; 69.0%; $p = 1.0$) with less than half the dimensions. Adding the six descriptors to the MFCCs gives the best handcrafted result (73.0%, 4.0 points above baseline: 35 errors corrected, 19 introduced; $p = 0.040$).

In the untuned setting of the progress report the same comparison gives +8.75 points ($p < 0.001$). The difference is mainly due to baseline tuning (64.8% → 69.0%) which improves about 4 points but combined set does not (73.5% → 73.0%). The tuned baseline is 3.7 points below what Piczak reported for a random forest on ESC-10 [3] (72.7%).

The pretrained deep reference is different league. With the same SVM YAMNet embeddings have 96.5% (macro-F1 96.5%), 23.5 points higher than combined set (101 vs 7 discordant clips, $p < 10^{-21}$)

and close to ESC-10 human accuracy reported [3] 95.7%. Every class has recall at least 90%, whereas best hand crafted model ranges from 62.5% (chainsaw) to 82.5% (crying baby).

Table 2. Clean ESC-10 results (400 out-of-fold predictions, official folds). Fold SD = standard deviation of the five fold accuracies.

Feature set	Dims	Untuned: C = 1, scale √26 (%)	Nested-tuned accuracy (%)	Macro-F1 (%)	Fold SD (points)
Temporal	4	56.3	63.0	62.5	5.4
Spectral	8	43.5	55.0	54.6	2.3
Descriptors (temporal + spectral)	12	60.8	69.3	69.3	2.1
MFCC baseline (MFCC + ZCR)	26	64.8	69.0	69.0	9.5
Combined (MFCC + descriptors)	36	73.5	73.0	73.0	8.0
YAMNet	2048	–	96.5	96.5	1.9

The cost comparison is less one-sided than accuracy. The handcrafted extractor (all 36 features) takes median 116 ms per clip in MATLAB and has no learned parameters. YAMNet has 3.75 million learned parameters (pretrained on about two million AudioSet clips), inference took only 56 ms per clip on the same CPU because it works on 16 kHz log-mel patches with optimised library code. The 36 handcrafted numbers can be read and checked directly; the 2048 embedding dimensions cannot.

4.2 The six background noises

The noises are very different (Table 3, Fig. 2). Car noise is almost pure rumble (99.99% below 500 Hz; LTAS centroid 26 Hz). Traffic and river noise are also low frequency dominated but the river noise is the most non-stationary (level SD 6.7 dB). Cafeteria and meeting-room noise contain speech babble and spectrally broader. Kitchen noise is only broadband noise with substantial high-frequency energy from cutlery and dishes (31% above 8 kHz; flatness 0.19).

Table 3. Characteristics of the DEMAND noises (channel 1, 300 s each).

Noise (DEMAND code)	LTAS centroid (Hz)	LTAS flatness	Power below 500 Hz (%)	Power above 8 kHz (%)	Level SD (dB)
Car interior (TCAR)	26	0.000	99.99	0.00	4.7
Street traffic (STRAFFIC)	464	0.025	91.7	2.6	4.1
Cafeteria (PCAFETER)	572	0.038	82.0	2.3	2.3
Kitchen (DKITCHEN)	3971	0.189	61.3	30.6	3.7
River (NRIVER)	109	0.006	96.5	0.2	6.7
Meeting room (OMEETING)	668	0.049	72.4	1.8	4.6

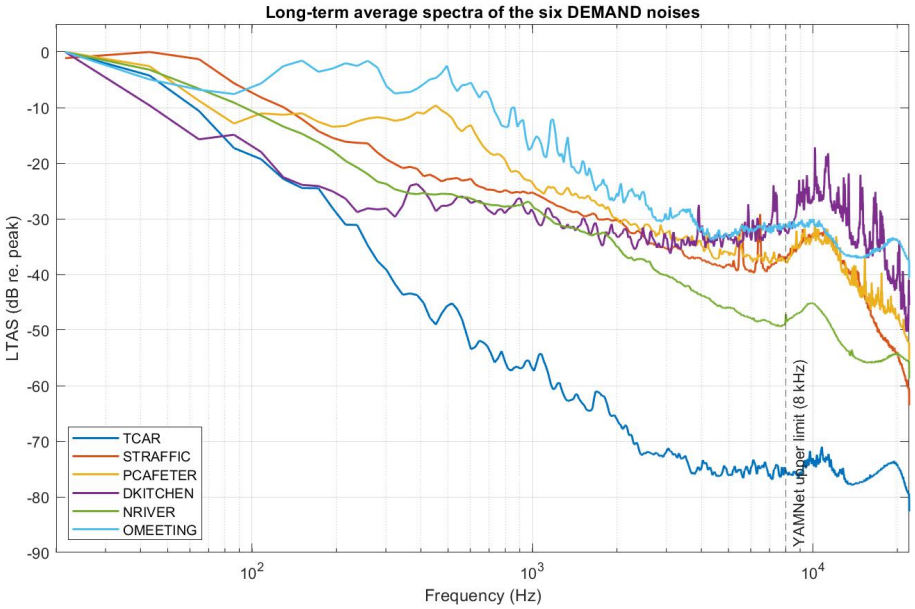


Figure 2. Long-term average spectra of the six noises, each normalised to its peak; the dashed line marks YAMNet's 8 kHz upper limit.

4.3 Robustness of clean-trained models

All models lose accuracy as the SNR falls, but the shape of the loss differs sharply between representations (Figure 3, Table 4).

- **The MFCC baseline collapses first.** At 20 dB, where the noise has only 1% of the event's power, the baseline already drops from 69.0% to 26.9% on average; at 0 dB it is at 14.0%, barely above chance (10%).
- **The interpretable descriptors degrade most gracefully among the handcrafted sets.** The descriptor set keeps 58.4% at 20 dB and 45.2% at 10 dB, and even the four temporal statistics keep 52.5% at 20 dB. The descriptor set is the best handcrafted set at every noisy SNR from 20 to 0 dB, with the temporal set close behind at 5 and 0 dB.
- **The ranking reverses.** The combined set, the best handcrafted set on clean data, falls to 41.8% at 20 dB, below both the descriptor and the temporal set, because it inherits the fragility of its MFCC part.
- **YAMNet is far more robust.** It keeps 86.3% at 20 dB and 63.6% at 0 dB on average, 66% of its clean accuracy against 44% for the descriptor set and 20% for the MFCC baseline. It beats the combined set in all 30 noise × SNR conditions (all McNemar $p < 10^{-5}$), but is still clearly affected: at 0 dB it loses 12.5 points in car noise and 45 points in river noise.
- **The noise type matters.** Low-frequency car noise is the least harmful noise for five of the six models (for the descriptor set, traffic is marginally less harmful). For the handcrafted sets that have not already collapsed, broadband kitchen noise is the most harmful (temporal 27.3%, descriptors 23.0%, combined 22.3% at 0 dB). For YAMNet the worst noises are river (51.3%) and cafeteria (54.8%), whose content resembles target classes (water, voices).
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Table 4. Mean accuracy (%) over the six noises for clean-trained models, and at 0 dB after multi-condition training on matched or unseen noise types.

Feature set	Clean	20 dB	10 dB	5 dB	0 dB	−5 dB	0 dB, multi-condition, matched	0 dB, multi-condition, unseen

Feature set	Clean	20 dB	10 dB	5 dB	0 dB	−5 dB	0 dB, multi-condition, matched	0 dB, multi-condition, unseen
Temporal	63.0	52.5	42.4	38.3	30.1	21.0	37.3	32.5
Spectral	55.0	41.2	32.8	26.2	22.0	17.8	41.0	30.5
Descriptors	69.3	58.4	45.2	38.4	30.3	22.0	50.4	42.9
MFCC baseline	69.0	26.9	20.7	17.3	14.0	12.3	55.8	42.9
Combined	73.0	41.8	32.0	28.8	25.3	22.8	66.8	53.8
YAMNet	96.5	86.3	78.9	72.8	63.6	51.9	87.6	78.1

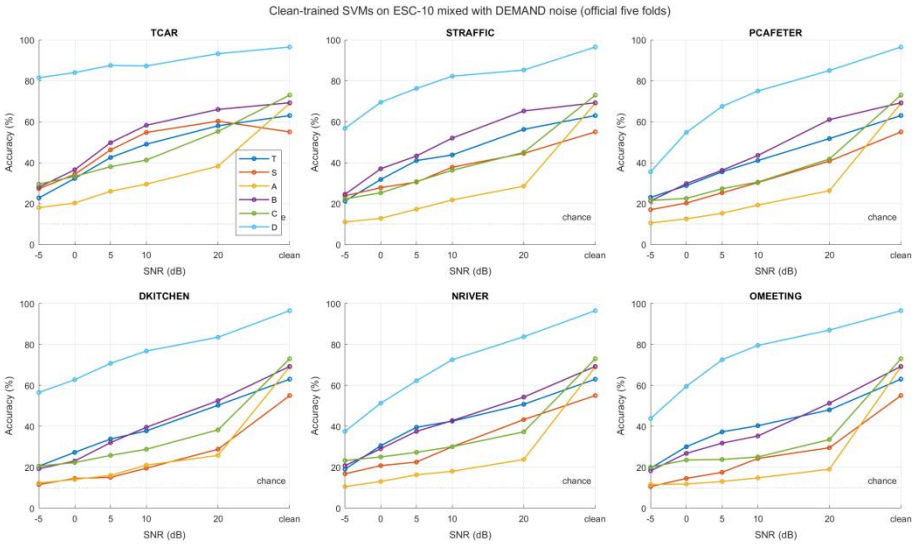


Figure 3. Accuracy versus SNR for every noise and feature set (clean-trained models; dotted line = chance). Legend: T temporal, S spectral, A MFCC baseline, B descriptors, C combined, D YAMNet; panel titles are the DEMAND codes of Table 3.

4.4 Why the models fail

MFCCs move about ten times further than the descriptors. Figure 4 shows how far each feature moves when noise is added (in units of its within-class SD; a shift of 1 is as much as the natural spread inside a class). At 20 dB MFCC means move by median 0.50 (car) to 1.25 (meeting room) within-class SDs but descriptor means only 0.03–0.18. For clean trained classifier one shift of one within-class SD can push clip into another class' region and 0.05 not. That's why MFCC baseline and combined set collapse at high SNR but descriptor and temporal sets do not.

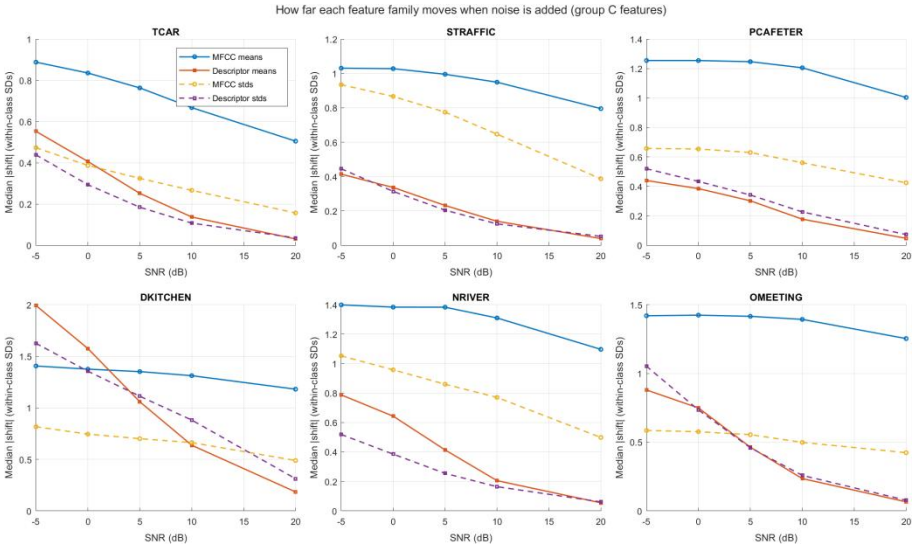


Figure 4. Median absolute paired shift of MFCC and descriptor statistics versus SNR, per noise (features of the combined set).

How the descriptors move follows the noise spectrum (Figure 5). Every noise increases mean RMS by about 1.2 within-class SDs at 0 dB (expected for added energy). Low frequency car noise pulls centroid, rolloff, spread and ZCR down (−0.36 to −0.44 SD). Broadband kitchen noise pushes them up strongly: spread +2.7; the standard deviations of rolloff and ZCR +2.1 and +1.9; flatness +1.6 SD. Babble and water noises raise mainly flatness and spread (+0.64 to +1.39 SD for river and meeting-room noise). The size of these shifts roughly matches how harmful each noise is for the descriptor based models: kitchen noise moves them most and is the most harmful.

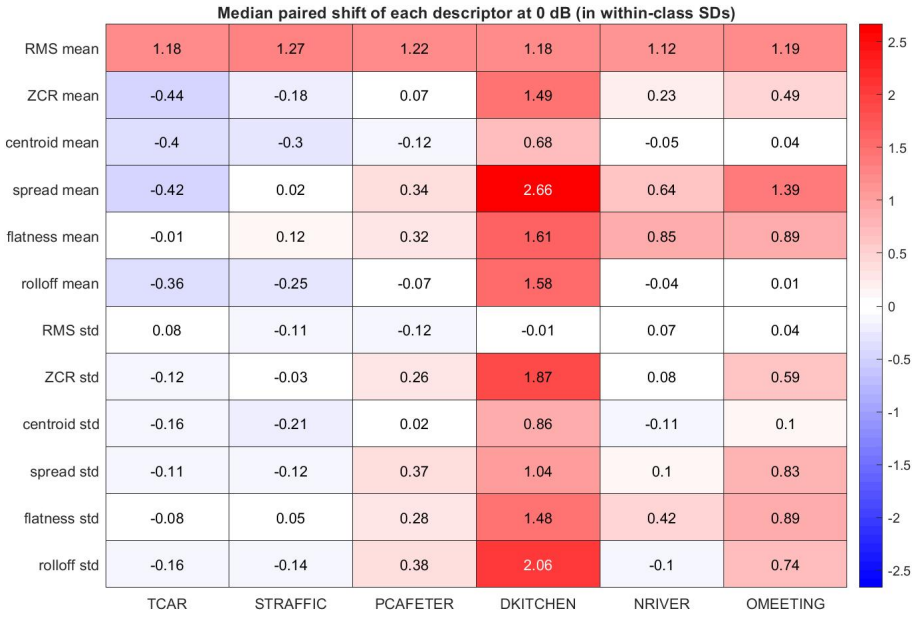


Figure 5. Median paired shift of each interpretable descriptor at 0 dB, in within-class SDs (red = up, blue = down; columns are the DEMAND codes of Table 3).

Noisy inputs are absorbed by a few classes. At 0 dB the MFCC baseline assigns 78–94% of all clips to clock tick in five of six noises; in kitchen noise 48% go to clock tick and 46% crackling fire. The combined set has the same tendency in weaker form (36–55% clock tick, 59% crackling fire in kitchen noise). YAMNet's errors are different kinds because they go to semantically related classes.

River noise sends 39% of clips to sea waves, car and traffic noise 20–22% to helicopter and kitchen and cafeteria clatter 33–36% to crackling fire.

Digital silence is a hidden cue. Clips in which more than 25% of frames are digital silence (mostly sneezing, rooster and dog) are much weaker. Of clips that were correct when clean 96% become wrong at 10 dB for temporal set, 86% descriptor set and 90% combined set against 27%, 31% and 59% of other clips. It happens for YAMNet too: 73% against 8% at 10 dB.

4.5 The modification: multi-condition training

Training on noisy copies changes the picture (Fig. 6, Table 4). When the test noise type was seen in training (matched), combined set mean accuracy at 0 dB goes from 25.3% to 66.8%, MFCC baseline from 14.0% to 55.8%. At 10 dB the combined set is above its clean accuracy (76.7% vs 73.0%).

The harder test is unseen where the test noise type is not used for training. The combined set still gets 53.8% at 0 dB (more than double clean-trained 25.3%) and YAMNet goes from 63.6% to 78.1%. For MFCC baseline, combined set and YAMNet every improvement at 0 dB is significant for every noise; for combined set with unseen noise gain is +17 to +40 points ($p \leq 4.1 \times 10^{-8}$).

The costs and limits are:

- **Clean accuracy:** the descriptor set, the combined set and YAMNet lose only 1–2 points on clean test clips (combined 73.0 → 71.3–71.6%; YAMNet 96.5 → 95.3–95.4%). The MFCC baseline and the temporal set lose more (69.0 → 59.5–60.6% and 63.0 → 58.8%).
- **Weak representations gain little:** the temporal set, with only four numbers per clip, improves by just 2–7 points at 0 dB.
- **The gap to YAMNet narrows but stays:** after multi-condition training, the combined set is again the best handcrafted set at every SNR (averaged over the noises), but it remains 20.8 points (matched) and 24.3 points (unseen) below YAMNet at 0 dB, down from 38.3.

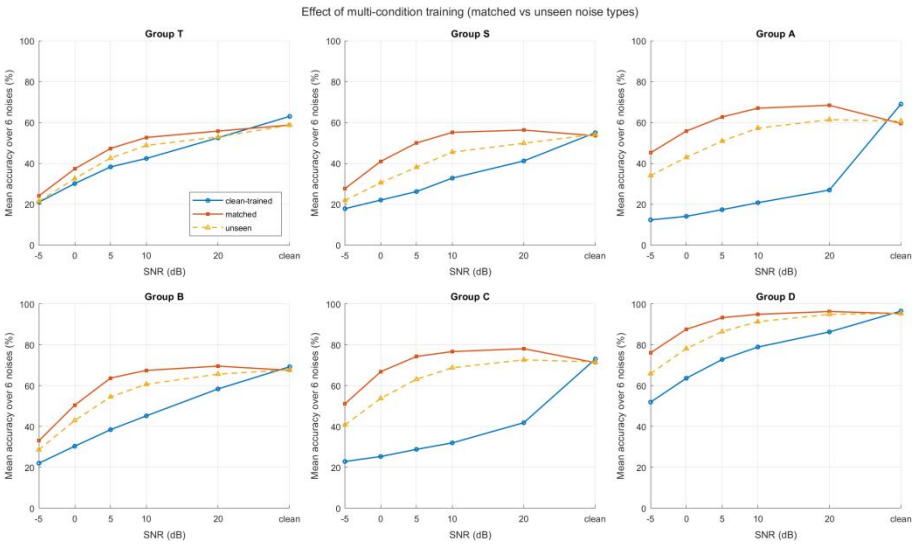


Figure 6. Mean accuracy over the six noises for clean-trained models and for multi-condition training with matched or unseen noise types. Panel titles: T temporal, S spectral, A MFCC baseline, B descriptors, C combined, D YAMNet.

5 Discussion

5.1 Answer to the research question

On clean data the interpretable descriptors are as good as MFCCs at half the dimensionality and together with MFCCs they give the best handcrafted result. Under real background noise a clean trained classifier based on these features is fragile (MFCC part fails even at 20 dB where a listener would hardly notice the noise, power-weighted descriptors degrade gradually).

How much damage a noise does is predictable. For the descriptors it follows the noise spectrum: broadband kitchen noise is worst and low-frequency car noise least harmful. For YAMNet it follows how much the noise resembles a target class. The pretrained deep network is far more accurate and robust but not immune to this pattern either. Multi-condition training (the modification tested here) recovers most of the loss for combined handcrafted set even for unseen noise types, but does not close the gap to the deep model.

5.2 Interpreting the results with signal-processing theory

Why MFCCs break first. MFCCs are the discrete cosine transform of the log mel-band energies [7]. Logarithm compresses large energies but expands small ones. Many ESC-10 clips have bands with almost no energy (above bandwidth of original recording) and frames of exact digital silence. A noise floor 20 dB below the event barely changes loud bands but lifts these near empty bands and silent frames by tens of dB on the log scale. That changes shape of log spectrum (and thus cepstral coefficients) about as much as differences between classes.

Spectral centroid, spread and rolloff do not behave like this. They are power weighted averages over the spectrum (strongest components dominate), so a weak additive noise moves them very little. This is what makes MFCC-based recognisers so bad in mismatched noise [9], [10]; here it is shown feature by feature.

Why the shift direction follows the noise spectrum. In the power domain the spectrum of a mixture is approximately the sum of the event and noise spectra. Power-weighted descriptor of the mixture is therefore pulled towards the noise's own value by proportion of the noise's share of power. Low frequency car noise pulls down the centroid; kitchen noise (31% above 8 kHz) pulls up. This additive view (which underlies Wiener filtering and spectral subtraction discussed in course) also explains the sign and size of the shifts in Figure 5.

Why noisy clips become "clock tick". With noise every clip has a broadband floor. Clock tick and crackling fire are sparse impulsive sounds on steady background; almost no digital silence (0.4% and 2.4% of frames). A noisy clip therefore looks more like clock-tick clip than its own class and the clean trained SVM puts it there. This is a failure of training distribution not of one feature so training with noise fixes it.

Digital silence as a dataset artefact. In clean ESC-10 the fraction of silent frames is itself informative. Sneezing, rooster and dog clips are partly padding so mean spectral descriptors pull to zero and SD features inflate. Noise fills in the padding and destroys that cue so clips with much silence flip far more often. This applies even for YAMNet whose embeddings are averaged over nine 0.96-s patches: short events get diluted by patches now containing only noise. The ESC-50 authors warn already about class-dependent preprocessing in the dataset [3] but these results show one concrete consequence. Part of what a model learns on clean ESC-10 is the recording format not the sound.

Why YAMNet is more robust and fails differently. YAMNet was trained on about 2 million YouTube clips that are themselves full of background noise and cover the ESC-10 classes among its

521 output labels [11]. Its representation has therefore seen sounds embedded in noise already. Log-mel input only covers 0–8 kHz so noise energy above 8 kHz is invisible to it (which partly explains why kitchen noise, hardest noise for 22 kHz handcrafted features isn't YAMNet's hardest). Its errors are semantic (river → sea waves, traffic → helicopter, clatter → crackling fire) because its embedding encodes sound categories rather than spectral shape.

YAMNet's advantage therefore comes from combining its representation with pretraining on data that the handcrafted models never saw, so it is not a comparison of architecture only. At the same nominal SNR YAMNet also gets different (band limited) SNR.

Why multi-condition training works, and where it does not. Multi-condition training does not make a feature less sensitive. It teaches the classifier where the noisy versions of each class lie in feature space (this is the standard explanation for its success in noisy speech recognition [16] and sound event classification [10]). It helps most for MFCC based sets whose clean trained decision regions were worst placed. Least for temporal set whose four numbers can't separate ten classes once noise has blurred them.

Two further observations matter in practice:

- **It generalises beyond the training noises.** Accuracy on unseen noise types is lower than on matched ones but far above clean training, so six varied noises are enough to teach the general effect of a noise floor.
- **The training mixture shapes the result.** The small loss in clean accuracy, and matched MFCC-based models doing better at 10–20 dB than on clean clips, show that the training distribution is now dominated by noisy data: six of every seven training examples are noisy. In deployment, this mixture should match the expected conditions.

5.3 Comparison with previous work

The clean handcrafted results are in line with Piczak's ESC-10 baseline [3], Chu et al. 's finding [5] that complementary time-frequency descriptors add to MFCCs; the fragility of clean trained MFCC systems in mismatched noise and large benefit of training on noisy data agree with Dennis et al. [9] and McLoughlin et al. [10]; here they are confirmed with real DEMAND recordings. YAMNet's 96.5% is consistent with gains reported for AudioSet-pretrained models on ESC-50 [13]–[15].

The study builds on that work in two ways. It compares interpretable descriptors, MFCCs and a pretrained network under identical paired noise conditions; and explains the differences at the level of individual features (log compression of near-empty bands and digital-silence artefact of ESC-10 are the main mechanisms).

5.4 Limitations and sources of error

- **Small dataset.** ESC-10 has only 40 clips per class, so per-class results are coarse and fold accuracy varies by several points. McNemar tests are not corrected for multiple comparisons, so the clean-data differences with $p = 0.005$ and $p = 0.04$ are weaker evidence than the noise results, whose p-values are below 10^{-5}.
- **Noise model.** Noise is added digitally from one microphone channel of six environments. Real recordings also include reverberation, microphone and gain effects. Six noises support a descriptive, not statistical, link between noise properties and degradation.
- **SNR definition.** The active-frame SNR (40 dB threshold) treats very quiet event tails as inactive; other thresholds were not tested.
- **Model choices.** The SVM grid is coarse, and several folds still choose the largest C. YAMNet is used frozen, not fine-tuned. Transformer models such as AST and BEATs, which lead on

ESC-50, were not tested; they could add little on clean ESC-10, but they may degrade differently in noise.
- **Fixed hyperparameters under multi-condition training.** They were kept at the clean-data choice to isolate the effect of the training data; tuning them on noisy validation data may improve the noise-trained models further.

5.5 Future work

- **Robust front ends:** floor or normalise the log-mel energies (for example per-channel energy normalisation), or apply the course's Wiener or spectral-subtraction front ends before feature extraction, to remove the log-compression sensitivity.
- **Silence-aware pooling:** compute clip statistics over active frames only, to remove the digital-silence artefact.
- **Stronger deep references:** fine-tune YAMNet, or test PANNs or AST with noise augmentation.
- **More data and conditions:** repeat the study on UrbanSound8K [2] with reverberation and microphone variation.

6 Conclusion

Interpretable temporal and spectral features are competitive on clean ESC-10 but weak under real background noise and a pretrained deep network is far more robust. On clean data four temporal statistics beat eight spectral ones (63.0% vs 55.0%), six interpretable descriptors match MFCC baseline (69.3% vs 69.0%), both together give 73.0%, YAMNet embeddings with same classifier reach 96.5%.

Under six real DEMAND noises clean MFCC features collapse even at 20 dB (26.9%). The analysis shows that log compression of near-empty bands and digital silence in ESC-10 move MFCCs about ten times further than power-weighted descriptors and the descriptors degrade predictably with noise spectrum. YAMNet is much better (63.6% at 0 dB) and its errors are semantically plausible.

The change tested (multi-condition training) more than doubles the 0 dB accuracy of the combined handcrafted set (25.3% → 66.8%). It still reaches 53.8% for noise types never seen in training at a cost of under two points on clean data. For real systems the lessons are: do not judge an environmental sound classifier on clean benchmarks only; train with realistic noise; use interpretable features to find out why a model fails.

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